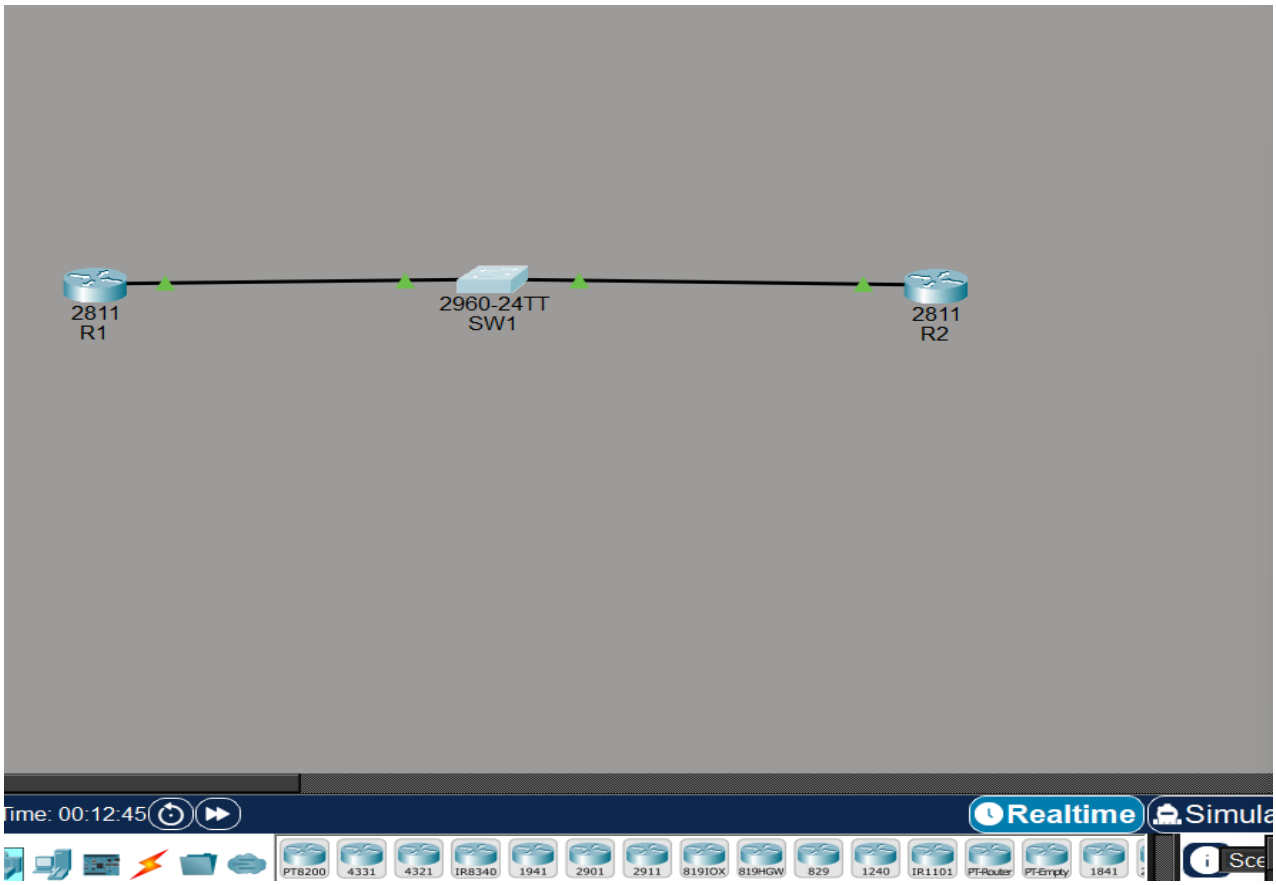
Week 5: Cisco Router and Switch Basics



1-8. Initial Configuration

R1 Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface fa0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#description Connection_to_SW1_Fa0/1
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#
```

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R2 Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface fa0/0
R2(config-if)# ip address 10.10.10.2 255.255.255.0
R2(config-if)#description Connection_to_SW1_Fa0/2
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]
R2#
```

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SW1 Configuration

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#interface vlan 1
SW1(config-if)# ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to
up

SW1(config-if)#ip default-gateway 10.10.10.2
SW1(config)#interface fa0/1
SW1(config-if)#description Connection_to_R1_Fa0/0
SW1(config-if)#interface fa0/2
SW1(config-if)#description Connection_to_R2_Fa0/0
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#write memory
Building configuration...
[OK]
SW1#
```

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Verify SW1 Gateway Connectivity

```
SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

SW1#
```

9. Verify Directly Attached Cisco Neighbours Using CDP

On R1:

```
R1>enable
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P -
Phone
Device ID     Local Intrfce    Holdtme     Capability    Platform     Port ID
SW1           Fas 0/0           124             S        2960         Fas 0/1
R1#
```

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On SW1:

```
SW1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P -
Phone
Device ID     Local Intrfce    Holdtme     Capability    Platform     Port ID
R1            Fas 0/1           136             R        C2800        Fas 0/0
R2            Fas 0/2           166             R        C2800        Fas 0/0
SW1#
```

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10. Prevent R1 from Discovering SW1 via CDP

Disable CDP on R1's interface connected to SW1:

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface fa0/0
R1(config-if)#no cdp enable
R1(config-if)#
```

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11. Flush the CDP Cache on R1

```
R1(config-if)#exit
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#
```

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12. Verify R1 Can No Longer See SW1

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P -
Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
R1#
```

13. Verify Status of Switch Port Connected to R2

On SW1:

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/1        unassigned      YES manual up
up
FastEthernet0/2        unassigned      YES manual up
up
FastEthernet0/3        unassigned      YES manual down
down
FastEthernet0/4        unassigned      YES manual down
down
FastEthernet0/5        unassigned      YES manual down
down
FastEthernet0/6        unassigned      YES manual down
down
FastEthernet0/7        unassigned      YES manual down
down
FastEthernet0/8        unassigned      YES manual down
down
FastEthernet0/9        unassigned      YES manual down
down
FastEthernet0/10       unassigned      YES manual down
down
FastEthernet0/11       unassigned      YES manual down
down
FastEthernet0/12       unassigned      YES manual down
down
FastEthernet0/13       unassigned      YES manual down
down
FastEthernet0/14       unassigned      YES manual down
down
FastEthernet0/15       unassigned      YES manual down
down
FastEthernet0/16       unassigned      YES manual down
down
FastEthernet0/17       unassigned      YES manual down
down
FastEthernet0/18       unassigned      YES manual down
down
```

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14. Shut Down the Interface and Verify

```
up
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to
administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed
state to down
```

Verify

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/1        unassigned      YES manual up
up
FastEthernet0/2        unassigned      YES manual administratively down
down
FastEthernet0/3        unassigned      YES manual down
down
FastEthernet0/4        unassigned      YES manual down
down
FastEthernet0/5        unassigned      YES manual down
down
FastEthernet0/6        unassigned      YES manual down
down
FastEthernet0/7        unassigned      YES manual down
down
FastEthernet0/8        unassigned      YES manual down
down
FastEthernet0/9        unassigned      YES manual down
down
```

15. Bring Interface Back Up and Verify Speed/Duplex

```
up
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed
state to up

SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show interfaces fa0/2 status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/2     Connection_to_R2_F connected    1          a-full  a-100
10/100BaseTX

SW1#
```

16. Set Duplex to Half



17. Restore Full Duplex

```
SW1#
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#duplex full
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0006.2a7e.8002 (bia 0006.2a7e.8002)
  Description: Connection_to_R2_Fa0/0
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
     Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 watchdog, 0 multicast, 0 pause input
     0 input packets with dribble condition detected
 --More--
```

18. Set Speed to 10 Mbps

```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#speed 10
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0006.2a7e.8002 (bia 0006.2a7e.8002)
  Description: Connection_to_R2_Fa0/0
 BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
     Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 watchdog, 0 multicast, 0 pause input
     0 input packets with dribble condition detected
 --More--
```

19. Check Interface on R2

On R2:

```
R2>enable
R2#show ip interface brief
Interface                  IP-Address      OK? Method Status
Protocol
FastEthernet0/0            10.10.10.2      YES manual up
up
FastEthernet0/1            unassigned      YES unset  administratively down
down
Vlan1                      unassigned      YES unset  administratively down
down
R2#
```

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Observation

Even though SW1 is operating at 10 Mbps, the interface on R2 generally remains operational because Ethernet devices can negotiate compatible speeds.

Reflection Question 20

When you set a duplex mismatch between SW1 and R2, what symptoms did you observe, and why does a mismatch cause this behaviour?

When SW1 was configured for **half-duplex** and R2 remained at **full-duplex**, the interface stayed **up/up**, but communication performance degraded significantly.

Symptoms Observed

- Slow network performance
- Intermittent packet loss
- Delayed ping responses
- CRC errors
- Late collisions
- Reduced throughput

Why This Happens

A duplex mismatch occurs when one side operates in **half-duplex** mode while the other operates in **full-duplex** mode.

- The full-duplex device can send and receive simultaneously.
- The half-duplex device expects only one transmission direction at a time and uses collision detection.

As a result, the half-duplex side detects collisions while the full-duplex side does not, causing retransmissions, frame errors, and poor overall performance. Although the link remains physically connected, data transmission becomes inefficient and unreliable.

Conclusion: The interface remains operational (**up/up**), but communication quality is significantly affected because the two devices are operating with different duplex settings.